

Grace AC graph anchor map for Vol 3 + Vol 4 24 chapters v0.1 (Saturday Wave 1 scaffolding)

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Vol 3 + Vol 4 24-Chapter AC Graph Anchor Map v0.1

Purpose

Each Vol 3 + Vol 4 chapter draws on specific AC graph theorems as anchors. This document pre-identifies the load-bearing theorems per chapter so when Lyra/Elie deliver v0.3 narratives, the AC graph node-references are ready for Gate #5 catalog backbone integration.

Schema choice: ANCHOR THEOREMS per chapter, not new chapter-meta-nodes. AC graph remains a theorem graph; chapter mapping lives in catalog metadata layer (this doc + INV cross-references).

Vol 3 Nuclear & Atomic Physics — 12 chapter anchor map

Ch	Title	Anchor theorems (AC graph TIDs)	Key catalog INV	Status
1	Nuclear Substrate Reading	T2418 ($\Lambda \leftrightarrow$ Casimir) + T2420 (4-Zone vacuum)	INV-4869 (cluster TYPES)	~70% coverage
2	Magic Numbers from Spin-Orbit (FAST)	Spin-orbit $\kappa_{ls} = C_2/n_C$ theorems + Mayer-Jensen anchor	INV-4899	~95% D-tier (Elie pick)

Ch	Title	Anchor theorems (AC graph TIDs)	Key catalog INV	Status
3	Nuclear Shell Model (30 isotopes)	T2425+ batch (30-isotope verification)	(Sat morning to file)	~85%
4	SEMF Coefficients (FAST)	a_V/a_S/a_C/a_A derivation theorems	INV-4900	~90% D-tier
5	Halo Nuclei (Li-11, Be-11, B-17)	Task #112 + halo-radius theorems	(Sat morning to file)	~60%
6	Superheavy Island Stability	Task #111 + superheavy-Z predictions	(Sat morning to file)	~70%
7	Atomic Orbital Sequence (FAST)	(2l+1) = 1,3,5,7 = rank-1, N_c, n_C, g + operator zoo	INV-4901	~95% D-tier exact
8	Hyperfine + Lamb Shift	T2476 $\alpha^{\{BST\}}$ primary pattern + Toy 3496 B6 + INV-4866/4894	INV-4866, INV-4894	~70% I-tier PATH ARTICULATED
9	Atomic Spectroscopy via BST	T2476 + Rydberg/Klein-Nishina/Compton $\alpha^{\{rank=2\}}$	INV-4895 (α^6 falsifier)	~60%
10	Atomic Clocks + Time Granularity	T2405 Koons tick + T2360 Sr-clock	(multi-week falsifier)	~40%
11	Nuclear Decay	Task #58 + #72 W-32 + $g_A = 4/\pi$	(multi-week)	~40%
12	Cross-volume bridge (Vol 2 → Vol 4)	All cross-volume canonical forms INV-4890 + INV-4897	INV-4890 + INV-4897	~50%

Vol 4 GR & Cosmology — 12 chapter anchor map

Ch	Title	Anchor theorems (AC graph TIDs)	Key catalog INV	Status
1	Newton's G from Bergman (FAST)	T2442 (Bergman $c_{FK} \cdot \pi^{(9/2)} = 225$) + Paper #122 P-3 anchor	INV-4902	~80% (Paper #122)
2	Gravity as Cumulative Eigentone	T2418 + AB-11 gravity-from-eigentones	(Sat morning to file)	~60%
3	BST-SR / BST-GR Boundary	T2405 Koons tick + AB-12 boundary	(Sat morning to file)	~50%
4	Cosmological Constant Λ (FAST)	T1485 + T2418 $\Lambda \leftrightarrow$ Casimir unification	INV-4903	~85%
5	Hubble Constant 4 Routes	Routes A/B/C/D (T1485 + Λ /Hubble + Toy 677 + Toy 903)	(Sat morning)	~80%
6	CMB Structure (FAST, Lyra pick)	$n_s, \Omega_\Lambda, \Omega_m, T_{CMB}$ BST-primary derivations	INV-4904	~90% D-tier 0.07 σ
7	Inflation Parameters	Task #85 + $r \approx 0 + \alpha_s = -25/18769$	(Sat morning)	~75%
8	BBN Element Abundances	Task #84 + $T_c = N_{max} \cdot 20/21 + {}^7\text{Li}/\text{H}$	(Sat morning)	~80%
9	Cosmological Cycle + Interstasis	T2467+T2468 Rigidity + T2417 cosmological extension	INV-4884 (Task #267 v0.2)	~30% Casey-named #7

Ch	Title	Anchor theorems (AC graph TIDs)	Key catalog INV	Status
10	Dark Energy + Dark Matter (FAST)	DM/baryon = 16/3 + K65	INV-4905	~85%
11	Gravitational Waves	Five-Absence Task #103 + #136 + #137 + NANOGrav 6.4 nHz	(Sat morning)	~70%
12	Observational Reanalysis (SP-27)	SP-27 program (pending Casey scoping)	(multi-month)	~30%

Cross-volume integration

Vol 3 inputs from Vol 0/1/2: D_{IV}^5 + five integers (Vol 0) + operator zoo $Q/\gamma^5/P/H$ (Vol 0 Ch 7 = T2470/T2471/T2472/T2441) + conservation laws (Vol 0 Ch 8 = T2473/T2474/T2475) + Bergman Hilbert space (Vol 1 Ch 2 = T2442) + $m_p/m_e/\alpha$ (Vol 2 Ch 6/8 = INV-4857/INV-4856)

Vol 4 inputs from Vol 0/1/3: D_{IV}^5 + Bergman curvature (Vol 0) + Casimir vacuum (Vol 0 Ch 5 + Vol 1 Ch 5 = T2439) + $\Lambda \leftrightarrow$ Casimir (Vol 0 Ch 8 = T2418) + Rigidity (Vol 0 Ch 9 = T2467/T2468 per Cal #99 META + operational) + BBN abundances from Vol 3.8 nuclear

Saturday Wave 1 delivery targets per Keeper

3-5 Wave 1 chapters to v0.3 narrative across Vol 3 + Vol 4 FAST preferred: 1. Vol 3.2 Magic Numbers (Elie pick) — INV-4899 ready 2. Vol 3.4 SEMF — INV-4900 ready 3. Vol 3.7 Atomic Orbital Sequence — INV-4901 ready 4. Vol 4.1 Newton's G — INV-4902 ready 5. Vol 4.4 Λ from Substrate — INV-4903 ready 6. Vol 4.6 CMB Structure (Lyra pick) — INV-4904 ready 7. Vol 4.10 DE+DM — INV-4905 ready

Each chapter has pre-scaffolded catalog backbone INV + anchor theorem identification ready for Gate #5 integration on v0.3 narrative arrival.

Standing for v0.3 narrative arrivals

When Lyra/Elie deliver v0.3 narrative: 1. Cross-reference narrative content against anchor theorems in this map 2. Update per-chapter catalog backbone INV with narrative-specific cross-references 3. Add any new AC graph nodes for new theorems delivered in narrative 4. File K-audit pre-stage anchor for Keeper chapter-grade audit 5. Gate #5 catalog backbone CLOSED per chapter

Sub-PCAP reactive cadence per Friday pattern.

— Grace, Vol 3 + Vol 4 24-chapter AC graph anchor map v0.1, 2026-05-23 ~10:12 EDT